

SELECT FINANCIAL CRISES BEFORE MID-1990s

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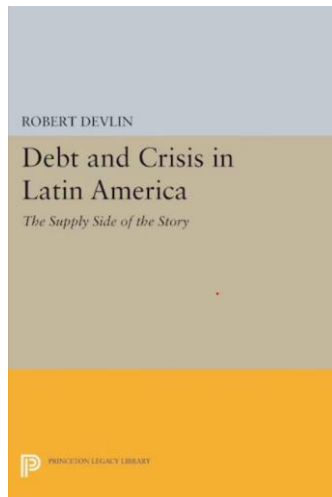
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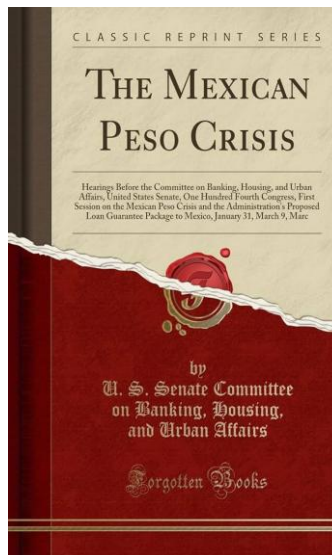
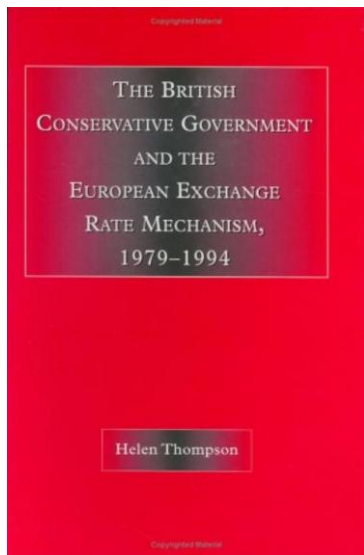
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Which Financial Crises will we Study in Today's two classes?

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- The Great Depression of 1929-39
- Latin American Debt Crisis of the 1980s
- The Exchange Rate Management (ERM) Crisis 1992-93
- The Mexican Peso (Tequila) Crisis of 1994-95



The Great Depression of the 1929-39

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Men throng to look for jobs reading notices by an employment agency during Great Depression



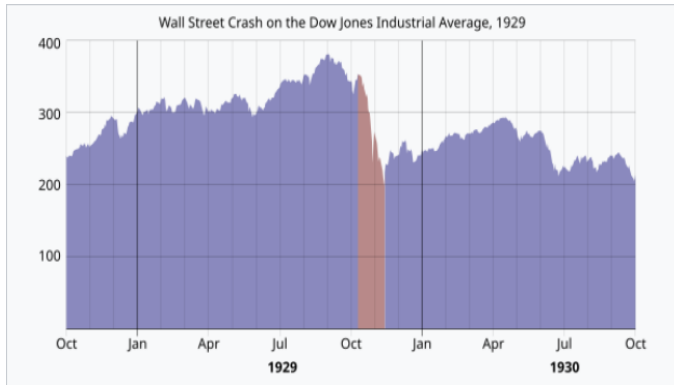
A line for bread at the 6th Av, 42nd St NYC during Great Depression



- The Great Depression has been the worst global economic downturn of the 20th Century that spilled over to many countries across the world.
- Its worst phase was during 1929-32, but the crisis lingered on till 1939
- It was on the back of the bubble built during “Roaring Twenties”, a decade in once the post-WW-I depression of 1920-21 ended, economic growth accelerated leading to widespread prosperity and spread of consumerism. Stock process also rose aided in part by consumption boom that was supported by feel-good factor was buffeted by art (e.g. art deco of visual arts, product designs and architecture) and music (e.g. jazz) flourished across several international cities (e.g. *années folles* or the crazy years in France). From 1926 to 1929, the market indices moved up nearly 400%.

The Great Depression of the 1929-39 was a crisis triggered by the stock market boom & burst, as has been the story of many other crises ... (1)

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Crowd gathers outside NYSE
after the Great Crash of 1929

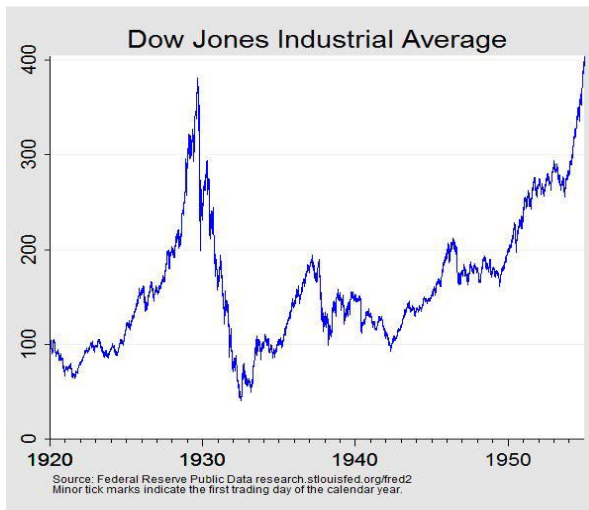
- October 24, 1929 (“Black Thursday”): The first market correction of overvalued stocks came on that day, when a record 12.9 million shares were exchanged, which was over 3X of the normal trading volume at that point. The systems for tracking the market prices could not keep up with trading volume and added to the panic selling. At one point, ticker tapes were running nearly 90 minutes behind the market. By the end of the day, the market had fallen 33 points or around 9%.
- October 28, 1929 (“Black Monday”): After a small recovery some recovery on October 25, Friday, many investors felt the market could rebound. However, some early profit taking negative sentiments returned and market confidence tanked resulting in high trading volumes with large sell-offs once again put pressure on the flow of information. Amid trading volumes were near 9.25 million shares, the market was down another 13%.

The Great Depression of the 1929-39 was a crisis triggered by the stock market boom & burst, as has been the story of many other crises ... (2)

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- October 29, 1929 (“Black Tuesday”): This is the day that most historians agree dealt the final blow to the Roaring 20s and was the starting point of the Great Depression. On Black Tuesday, a record 16.4 million shares exchanged hands. The ticker tape machines fell behind by nearly 3 hours. With all hope of a market recovery now gone, panic selling continued and the market fell another 12%.
- Recovering from the 1929 Stock Market Crash: Over the next month the market continued to decline sharply. The market bottomed out only in July 1932, when the Dow hit 41 from a high of 381 in 1929. Overall, the market declined by nearly 90%. Even as the market started to rise in 1932, it would take another 22 years before the Dow would climb above the 1929 levels.



The Great Depression left deep scars on the US & European Economies during 1929-32:

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- **Great Depression (1929-33/ 1929-39):** It started with the Stock Market boom of 1920s that went bust in 1929 leaving deep scars on economy. The worst was over by 1933 but after effects continued till 1939. It has been the biggest, & the longest financial crisis ever that resulted in the deepest recession ever in the history
- Changes in major macro-parameters 1929-32:

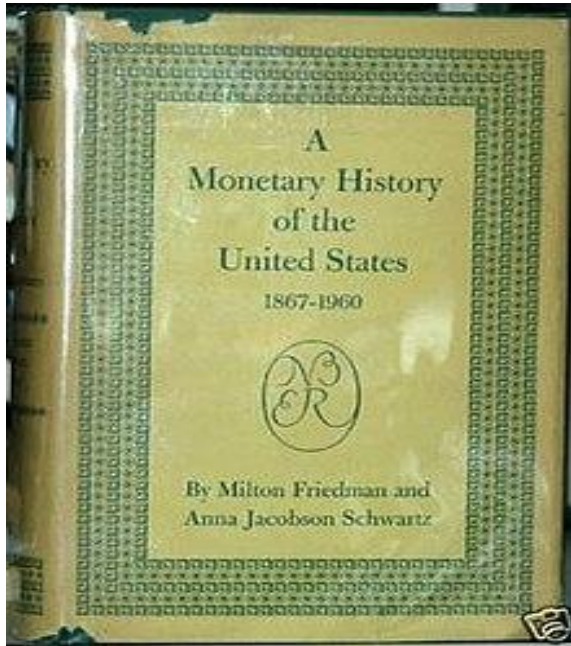
	US	UK	France	Germany
Industrial production	-46%	-23%	-24%	-41%
Wholesale prices	-32%	-33%	-34%	-29%
Foreign trade	-70%	-60%	-54%	-61%
Unemployment	+607%	+129%	+214%	+232%



- US unemployment rate rose from 3.2% at the time of market crash to 24.9% in 1933; GDP contracted 8.5% in 1930 and 12.9% in 1932; inflation turned negative with deflation rates of 9.3% in 1931 and 10.3% in 1932

Milton Friedman & Anna Schwartz on the Causal factors Behind the Great Depression

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The Great Depression-related chapter, "The Great Contraction", was republished as a separate book in 1965 focussing on 1929-33 period; was reprinted by Princeton Univ Press in 2008.

Friedman and Schwartz identified four main policy mistakes made by the Fed that led to a sharp and undesirable decline in the money supply:

- In the spring of 1928, the Fed began to tighten its monetary policy (resulting in rising interest rates) and continued that same policy until the stock market crash of October 1929. It was this tight money policy that caused the economy to enter a recession in mid-1929 and triggered the stock market crash a few months later.
- In the fall of 1931, it raised interest rates again to defend the dollar in response to speculative attacks, ignoring the difficulties this caused to domestic commercial banks.
- After lowering interest rates early in 1932 with positive results, it raised interest rates again in late 1932, causing a further collapse in the U.S. economy.
- The Fed was also to be blamed for a pattern of ongoing neglect of problems in the U.S. banking sector throughout the early 1930s. It failed to create a stable domestic banking environment by supporting the domestic banks and acting as lender of last resort (LOLR) to domestic banks during banking panics.

Milton Friedman & Anna Schwartz Deep Dive on the Great Depression

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Chapter 7 section 7 they starkly ask the question "Why was Monetary Policy so Inept?"... & then answered it as follows: "The explanation ... is the shift of power within the System and the lack of understanding and experience of the individuals to whom the power shifted." The truth is that the demise of Benjamin Strong, the Fed NY President, due to tuberculosis in October 1928, left Fed Reserve Board very weak.

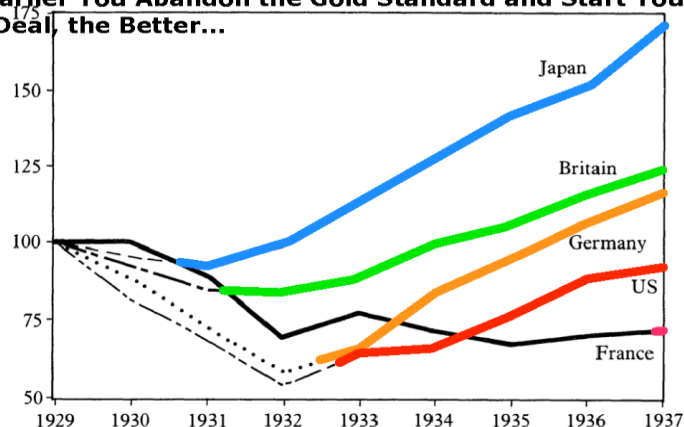
- Milton Friedman and Anna Schwartz were at the working at the National Bureau of Economic Research (NBER) when Arthur Burns, who became a Fed Chair later, suggested that the two could collaborate on a project to analyze the effect of the money supply on business cycles. Schwartz had already gathered important historical data at that point, while Friedman had earned reputation as a monetarist being a Professor at the University of Chicago.
- They produced one of the most influential book 'A Monetary History of the United States 1867-1960', that was published by the Princeton University Press. The book argued, then a novel proposition, that changes in the money supply profoundly influenced the U.S. economy, especially the behavior of economic fluctuations (business cycles). The message from the book was that changes in the money supply had unintended adverse effects, and that sound monetary policy is necessary for economic stability.
- The chapter in this book dealing with the causes of the Great Depression was later also published as a stand-alone book titled, "The Great Contraction, 1929–1933".

Barry Eichengreen attributes the Fed's rate hikes to the Gold Standard

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THE GREAT SLUMP REVISITED

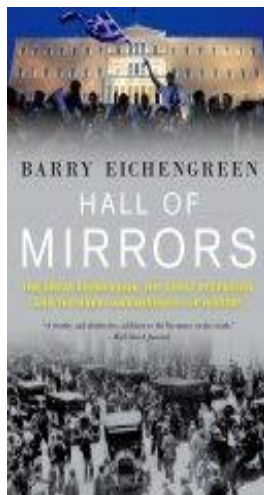
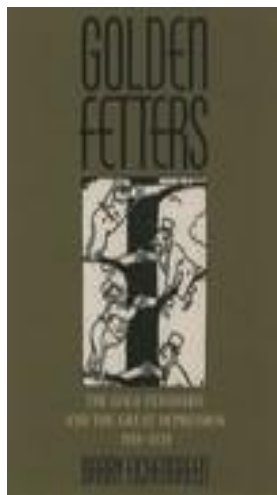
The Earlier You Abandon the Gold Standard and Start Your New Deal, the Better...



Eichengreen (1992), "The Origins and Nature of the Great Slump Revisited"
Figure 5. *Indices of industrial production, 1929-1937* (1929 = 100)
Source: League of Nations, *World production and prices, 1937/8*, p. 44.

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- The three discount rate hikes in 1920, 1931 and 1937 by the Fed contributed to Great depression. Each of these actions caused a large contraction of the money supply over the next 12 months (9%, 14%, 3%). In fact, these were the three sharpest declines Friedman & Schwartz they saw in their data. In those 12 months, industrial production fell dramatically (30%, 24%, 34%).
- But why did Fed raise discount rate? Barry Eichengreen has argued that because of the gold standard the hands of Fed were tied. In order to maintain the credibility of the gold standard, it could not undertake actions to drastically increase the money supply in the manner advocated by Friedman and Schwartz.
- As euro crisis prevailed during 2009-10, Eichengreen & Temin (2010) argue that gold standard and the euro, were both extreme forms of fixed exchange rates and were bad ideas. While the world was lucky to have avoid another Great Depression in 2008-09 due to aggressive monetary and fiscal stimuli, problems could intensify in bad times.



Great Depression was a monetary shock propagated by the Fed

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"I would like to say to Milton and Anna: Regarding the Great Depression. You're right, we did it. We're very sorry. But thanks to you, we won't do it again.

—Ben Bernanke, November 8, 2002, in a speech "On the Occasion of His 90th Birthday." at a Conference to Honor Milton Friedman.

- The **First episode**: the deliberate tightening of monetary policy that began in the spring of 1928 and continued until the stock market crash of Oct. 1929. As Friedman and Schwartz noted, the business-cycle trough had only just been reached at the end of 1927 (the NBER's official trough date is Nov. 1927), commodity prices were declining, and there was not the slightest hint of inflation. Why then did the Fed tighten in early 1928? Bernanke states a principal reason was the Board's ongoing concern about speculation on Wall Street and for long its distinction between "productive" and "speculative" uses of credit.
- The **second episode**: In Sept. 1931, following the sterling crisis, a wave of speculative attacks on the pound forced Great Britain to leave the gold standard. Anticipating that the U.S. might be the next to leave gold, speculators turned their attention from the pound to the dollar. The resulting outflow of gold reserves also put pressure on the U.S. banking system, as foreigners liquidated dollar deposits for gold. FRBNY raised its rediscount rate to 2½%, and on Oct. 16 to 3½% and eventually to 5% - the sharpest rise in a short period. This action stemmed the gold outflows but 522 commercial banks had to close in Oct. alone.
- The **third episode**: started with contractionary impact in the period from Jan. 1933 to the banking holiday in March. This was the period of uncertainty before President elect Hoover could take office. Markets read President-elect's campaign statements correctly and inferred that Roosevelt might devalue the dollar or even break the link with gold entirely. Fearing the resulting capital losses, both domestic and foreign investors began to convert dollars to gold, putting pressure on both the banking system and the gold reserves of the Fed. More bank failures followed & the economy took its deepest plunge between Nov. 1932 and March 1933

The Latin American Debt Crisis of the 1980s

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- The routes of the crisis lay in the two large oil price shocks in the 1970s that caused widening of current account deficits in many Latin American (LatAm) countries. Oil-exporting countries, however, ran large current account surpluses that got parked in US banks and financial markets and then got on-lend to LatAm countries. Throughout the 1970s, Latin American nations, particularly Brazil, Argentina, and Mexico, relied heavily on foreign commercial bank loans to fund development and cover rising oil prices
- Total outstanding debt from all sources ballooned from only US \$29 billion (bn) in 1970 to US\$159 bn by the end of 1978 and further to US\$327 bn. The crisis sparked by Mexican FM Jesús Silva Herzog informing Fed Chair, US Treasury secretary & IMF MD in Aug. 1982 that Mexico would no longer be able to service its debt, which at that point totalled US\$80 bn



The Latin American Debt Crisis of the 1980s

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- In October 1979, Fed Chairman Paul Volcker announced new measures by the Federal Open Market Committee (FOMC) aimed at reining in the inflation. Inflation in the US surged from 2.8% in 1967 to 14.8% in March 1980. Tight monetary policy raised Federal funds rate from an average of 11.2% in 1979 to 20.0% in March 1980, also setting off monetary policy tightening even across Europe and elsewhere. This caused a severe global economic downturn. With high US interest rates, international banks cut down their lending to LatAm, moving money into US assets. Consequently, the LatAm debt bubble burst. Mexico was the first to default, but many other LatAm countries followed.
- The crisis was eventually alleviated when the U.S. announced **the Brady Plan on 11th April 1989**, which allowed for the reduction and restructuring of commercial debt into tradable bonds. 18 countries agreed to Brady deals forgiving \$60 bn of debt in about \$190 bn in bank claims
- Initially, the Brady Plan was dismissed as **“an offer to sell fire insurance at bargain rates in a town where half the people are arsonists.”** (Cato Journal, 16(2), Fall 1996)



Nicholas Brady

Nicholas Brady, Treasury Secretary
(under Reagan & later under Bush).

The Policies of heterodox shocks...(1)

What were these policies?

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- Stabilization programs introduced by Argentina, Bolivia and Brazil (also in Israel) between July 1985 and March 1986, when hyperinflation was an imminent threat to each of these economies, so that a primary policy aim had to be the attainment of a quick and drastic reduction in inflation.
- However, inflation in Latin America was “inertial” (implying automatic move of the inflation in the same direction) in characteristic, meaning that with high level of indexations in wage and other contracts (wages were fully indexed in many cases, implying nominal wages automatically increased in line with inflation rate to compensate labour fully)
- They introduced heterodox shocks. *First*, in an effort to influence expectations they created major discontinuities in the trends of the main macroeconomic variables, rather than following the gradual adjustment paths of traditional stabilization packages. For this reason, they have usually been referred to as “shock” programs. *Second*, in most cases, they included a number of incomes-policy elements, such as wage and price freezes, together with exchange rate pegging and deindexation measures. These components have been characterized as “heterodox,” as compared with the “orthodox” emphasis on demand management through fiscal and monetary instruments.
- However, in practice, traditional “fundamental” policies, mainly in the monetary and fiscal areas, formed the bulk of each program. In fact, the designers of the shock programs considered the combination of heterodox and fundamental (demand-management) policies crucial to the success of the programs.

The Policies of heterodox shocks: The Case of Brazil... (2)

freezes, controls, and
incomes policies are
not costless. T



Often, an important components of these heterodox shock policies was to replace one currency with another. Since 1942, Brazil replaced currencies seven times (had eight different currencies, starting with the Cruzeiro).

- **The Cruzado Plan (1986):** Launched by President José Sarney, it replaced the *cruzeiro* with the *cruzado* (at 1,000 cruzeiros = 1 cruzado). It was imposed with immediate general price and wage freeze, & de-indexing the economy.
 - **The Outcome:** The plan created a temporary surge in consumer purchasing power, causing widespread shortages and a massive wage-price spiral of demand-induced inflation. By late 1986, the plan collapsed. Inflation rebounded.
- **The Bresser and Summer Plans (1987 & 1989):** As the inflation had re-accelerated, the government introduced the Bresser Plan (1987) and the Summer Plan (1989), both featuring new price freezes and temporary de-indexation.
 - **The Outcome:** The heterodox measure offered only temporary relief. Price controls quickly became unsustainable as firms anticipated future freezes and created supply distortions.
- **The Collor Plans (1989-91):** President Fernando Collor de Mello introduced sweeping heterodox measures accompanied by aggressive orthodox policies. The cruzado novo was briefly introduced exchanging 1000 cruzeiro=1 cruzado novo but it proved to be short-lived only for the period 15th Jan, 1989-15th Mar 1990. The currency was later renamed cruzeiro. The government also shockingly confiscated large fractions of private savings and checking accounts to restrict liquidity.
 - **The Outcome:** The asset seizures caused loss of consumer & business confidence resulting in a deep economic contraction and widespread hardship, yet failed to sustainably break the country's inflationary inertia.



The Policies of heterodox shocks:

The Case of Brazil - Plano Real ... (3)

freezes, controls, and
incomes policies are
not costless. T

- **The Plano Real (1994):** Was designed by a team of economists including President Fernando Henrique Cardoso himself. This program departed from prior heterodox shocks by focusing on de-indexing rather than enforcing immediate price freezes. In March 1994, the government introduced the Unidade Real de Valor (URV) as a transitional unit of account tied to the U.S. dollar. Prices, wages, and other contracts were slowly converted into URVs, which stabilized inflation expectations. This allowed Brazilians to recalibrate contracts and compare prices in a stable unit while still transacting in the old currency, the *cruzeiro real*, which at that point had circulated for only 10 months. On July 1, 1994, the government officially introduced the **Brazilian Real (BRL)**, replacing the *cruzeiro real* at parity with the URV. Because the transition period already de-indexed the economy, the introduction of the Real did not trigger the same shock-induced demand or supply crunches seen in 1986.
- **Outcome:** Supported by tight monetary policies and fiscal adjustments, it succeeded in finally taming Brazil's chronic hyperinflation. From over 2,000% annual inflation rate dropped to single digits in 1996. But the gains of this successful disinflation were frittered away by lack of fiscal discipline. Long-term debt rose again causing a currency crisis in Brazil in 1999. The Real initially appreciated against the USD led by large capital inflows in late 1994 and 1995. It then began a gradual depreciation, culminating in the Jan. 1999 free fall, when the real suffered a maxi-devaluation amid wide fluctuations. The period of a quasi-fixed exchange rate during 1994-1999, was succeeded by Brazil central bank president Armino Fraga adopting inflation-targeting, which effectively meant that the fixed-exchange period was over. However, the currency was never freely floating but a "dirty" float, with frequent central bank interventions to manipulate its dollar price. Since Brazil's macroeconomic stabilization program was implemented in 1994, the percentage of Brazilians living in poverty has decreased and the distribution of income has registered a small improvement. But poverty and income inequality remain severe.

The Policies of heterodox shocks:

The Case of Brazil: the 1999 Currency Crisis & the Samba Effect... (4)

freezes, controls, and
incomes policies are
not costless. T

- **The Samba effect (1999):** nickname for the currency crisis in Brazil; Real depreciated by 35%. Three main causes:
- **(1) Dirty Float:** While Brazil successfully ended hyperinflation by pegging the new currency (the real) to the US dollar as part of the Plano Real, it continued to try keeping new currency stable and the exchange rate regime worked like a Dirty Float. Therefore, real became overvalued over time, causing cheap imports to flood the country and leading to massive trade deficits;
- **(2) Contagion:** Spillovers from global shocks hit the Brazilian financial markets. The 1997 Asian Financial Crisis increased country risk premia for EMEs steeply widening spreads. As Asian crisis sparked global slowdown, demand for oil fell severely affecting the fiscal position of Russia, which is a major oil exporter. Public Debt strains magnified at a time when political crisis brewed as in 23rd March 1998, President Boris Yeltsin suddenly dismissed PM Viktor Chernomyrdin & his entire cabinet & named 35-year old Energy Minister Sergei Kiriyenko as acting PM. The Russian GKO crisis that caused Russian government to default on its sovereign debt and devalue Ruble on 17th August 1998 had ripple effects on many EMEs, including Brazil; GKO's were short-term, zero-coupon Russian T-bills on which Russia government was offering super-normal yields
- **(3) High Interest Rates:** To defend Ruble and stem the capital flight, Brazil's central bank resorted to interest rate hikes as a line of defence raising short-term rates to extremely high levels. This made it very expensive for the government to pay off its debt and sparked fears of a default. The Collapse followed.

Conclusion: Policies of heterodox shocks have succeeded on some occasions but miserably failed on others; so difficult to make a virtue of these policies. Direct control or freezes of prices have at times delivered reduced inflation but are not costless as markets don't clear, leading to real output losses and inefficient allocation of resources. It is best to use heterodox policies sparingly when alternative policies to dampen inflation expectations are not working & complement them with a strong structural adjustment programme.

What were Brady Bonds? How Did They Work?

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- Brady bonds, designed to help LatAm & other EM countries to restructure their debt, were 25-30 year USD-denominated debt securities issued by developing countries. They were collateralized by zero-coupon US T-bonds of the same tenor. The bonds got popularly named after former U.S. Treasury Secretary Nicholas Brady, who took this initiative, now called the Brady Plan. The funds required to collateralize and securitize these restructured bonds came from the IMF and the World Bank (\$12bn each), and the Japanese Import-Export Bank \$8 bn. Banks accepted average write down of 30-35%. Brady bonds opened the access to international capital markets for EME countries reeling under huge public debt. Creditworthiness of the issuing countries went up and their debt became more liquid as it became attractive to international investors. But, the EME governments issued these bonds only sporadically and on a limited basis through most of the 20th century.
 - Interest payments on Brady bonds were based on a fixed or variable rate, offering investors different levels of risk and return. They were issued either as par bonds (with below market rates) or deep discount bonds (with 35-50% haircuts) or Front-Loaded Interest Reduction Bonds (FLIRBs). They were mostly denominated in U.S. dollars, but some small issuances were also done in German marks, French and Swiss francs, Dutch guilders, Japanese yen, Canadian dollars, and British pounds. The long-term maturities of Brady bonds make them attractive vehicles for profiting from spread tightening.
- Argentina
 - Brazil
 - Bulgaria
 - Costa Rica
 - Cote d'Ivoire
 - Dominican Republic
 - Ecuador
 - Jordan
 - Nigeria
 - Panama
 - Peru
 - The Philippines
 - Poland
 - Russia
 - Uruguay
 - Venezuela
 - Vietnam

Q: Think as an investor: Do Brady Bonds carry any risks or would you consider them as safe assets?

The European Exchange Rate Mechanism (ERM) Crisis... (1)

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- There have been on and off formal exchange rate arrangements made between European countries between “ins” (those who want integration/ union) and “outs” (those who want to stay out of such arrangements).
- The **European Exchange Rate Mechanism** (now known as **ERM I**) was an arrangement established by the European Community (EC) in 1979 as a system of adjustable pegs that lasted till 1st Jan. 1999 when the single currency euro was launched. The system also introduced **European Currency Unit (ECU)** as a composite "artificial" currency based on a basket of European Union (EU) member currencies, weighted according to each country's share of EU output. The ECU started off with nine currencies and by late 1989 through the end of 1999, the basket held 12 currencies. The currencies were the Belgian franc, German mark, Danish krone, Spanish peseta, French franc, British Pound, Greek drachma, Irish pound, Italian lira, Luxembourg franc, Dutch guilder, and Portuguese escudo.
- While it initially operated to reduce exchange rate variability in the floating rate era, the foundational blueprint for a **three-stage transition** to euro was laid by the Delors Report in 1989 named after EC President Jacques Delors, that outlined a step-by-step transition to a single currency. Stage-I from July 1990 to 1993 aimed at market integration through complete liberalization of capital movements amongst EEC states. Stage-II from 1994 to 1998 aimed at economic and institutional convergence by enhancing independence of the European National Central Banks, enhancing fiscal discipline and setting up European Monetary Institute (EMI) as a precursor to European System of Central Banks (ESCB). Stage-III was to launch euro as a single currency in the euro area comprising of those countries who met the **Convergence Criteria** by irrevocably fixing the exchange rates of these countries with the euro.

The European Exchange Rate Mechanism (ERM) Crisis... (2)

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- The ERM crisis occurred as currency crisis in 1992-93 when ERM-I had actually collapsed even in its first stage. Economic fundamentals (e.g., price levels) had diverged substantially for the ERM countries, and a fixed exchange rate could not be sustained and ultimately came under speculative attack. This view has found a theoretical expression in so-called first-generation models of speculative attacks (Krugman, 1979, and Flood and Garber, 1984). An alternative story stresses the fragility of fixed exchange rate regimes because of the self-fulfilling nature of speculative attacks. This story has been given a strong theoretical foundation in so-called second-generation models of speculative attacks (Obstfeld, 1986 and 1995). There is a broad consensus that the collapse of ERM I can be seen as a combination of these two stories. There is less consensus, however, on the relative importance given to these two explanations.
- However, ERM was stitched back
- The **Convergence Criteria** was laid down in the **Maastricht Treaty** of 1991.

The European Exchange Rate Mechanism (ERM) Crisis... (2)

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The **Convergence Criteria** was laid down in the **Maastricht Treaty** of 1991.

- **Low inflation.** The average inflation rate observed during a one-year period before a country is examined for admission to the single currency must not be more than 1.5% higher than the average of the three best performing Member States in terms of price stability.
- **Sound public finances.** The government deficit must not exceed 3% of gross domestic product (GDP) and the public debt must not exceed 60 % of GDP, unless the ratio is sufficiently diminishing and approaching the reference value at a satisfactory pace. This latter criterion is therefore more flexible with greater scope for discretion.
- **Stable exchange rates.** Candidate countries must have observed the normal fluctuation margins provided for by the exchange rate mechanism of the European Monetary System for at least two years, without devaluing their currency against that of any other Member State.
- **Low interest rates.** During the year preceding the examination, the average long-term interest rate must not be more than 2% above that of the three best performing Member States in terms of price stability.

The European Exchange Rate Mechanism (ERM)-II... (2)

<https://www.youtube.com/watch?v=8qYeaTBC48k>

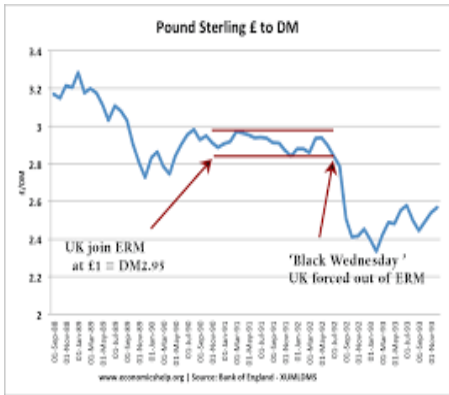
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- The **European Exchange Rate Mechanism II (ERM II)** was established as part of the Maastricht Treaty in 1999 to guide the entry into **European Monetary Union (EMU)** with single currency euro for those who may wish to join it in future (after the launch of euro). It seeks to minimize their currency volatility and serves as a mandatory stepping stone for EU member aiming to officially adopt the euro as their national currency. Successful participation in ERM II is part of the 'convergence criteria' set for entry into EMU.
- **Central parity:** When a country joins ERM-II, its central bank and the ECB agree on a central exchange rate against the euro. Central parities under ERM-II can be adjusted to prevent significant misalignments provided all parties agree in a confidential procedure.
- **Fluctuation Band:** The national currency is then allowed to fluctuate within $\pm 15\%$ around this central parity.
- **Intervention:** If the currency drifts too close to its upper or lower limits, the participating central banks intervene (e.g., by buying or selling the currency) to stabilize it within the allowed range. The commitment of the ECB to intervene in the FX market to support a particular non-euro currency is agreed to be limited under ERM-II.
- **Countries under ERM II:** Set up in 1997, the Danish krone was the only currency participating in ERM II for several years. After 2004 enlargement of the EU, the currencies of the Baltic countries, Cyprus, Malta, Slovakia and Slovenia all entered the mechanism. Croatian kuna and Bulgarian lev entered ERM II in 2020 & adopted euro in 2023 & 2026, respectively. Denmark had joined ERM-II in 1999 but opted out due to "No" referendum on Maastricht Treaty & later on euro. But it remains part of ERM II & ties itself to $\pm 2.25\%$ fluctuation band to euro.

The European Exchange Rate Mechanism (ERM) Crisis... (3)

<https://www.youtube.com/watch?v=sNEmqR-gNX4>

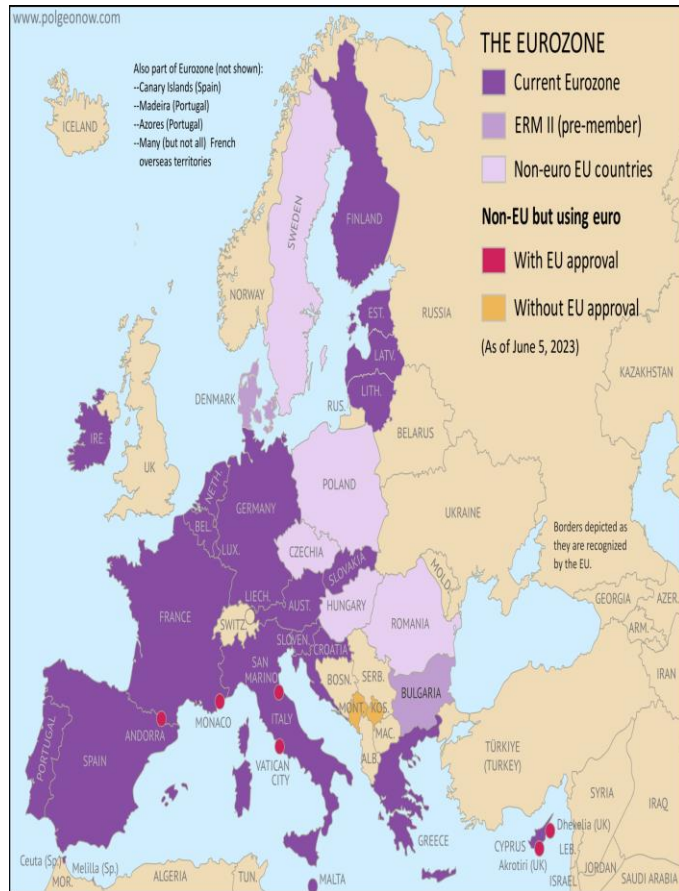
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- In 1990, UK formally entered ERM at a parity of 1GBP=2.29DM. ERM allowed $\pm 2.25\%$ movements around the central rate.
- ERM was malfunctioning. UK, as most other ERM countries except Germany and some smaller countries, were repeatedly failing to keep “snake within the tunnel” that prompted magazine Economist to call it a “Rattlesnake”.
- The 1992 sterling crisis peaked on “Black Wednesday” on 16 September 1992 when the UK government was forced to withdraw sterling from the ERM as it failed to keep its exchange rate above the lower limit required under the ERM. it was embarrassing as at that time, the UK held the Presidency of the Council of European Union (EU).
- In the run up to the Black Wednesday, George Soros through his Quantum Fund and some other currency traders had built huge short positions in sterling betting on the collapse of Sterling below the lower band of the ERM. On Black Wednesday he shorted about sterling £10 bn. Despite BoE making FX selling interventions that was said to have crossed £2 bn an hour at one stage and estimated to have cumulatively costed £22 bn in the run up to that Wednesday, sterling collapsed. BoE got defeated by Soros. Official estimates place the loss for BoE at £3.3 bn. Calculations showed that if BoE had maintained its £24 bn of FX reserves and sterling would have fallen the same amount as it did, BoE would have made profits of £2.4 bn. Bank of England has stopped intervening in FX markets after the debacle.

How the ERM Crisis Unfolded?

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Euro area as it is now

- When the ERM was set up in 1979, UK had refused to join. This was despite the Chancellor of Exchequer, Geoffrey Howe being a staunch European supporter. His successor Nigel Lawson was an advocate of fixed exchange rate system. But UK joined ERM only in 1990 and left in 1992 obtaining opt out from joining EMU in return of agreeing to sign Maastricht Treaty, which it did 1992 after securing two opt outs – (i) from euro and (ii) from Schengen.
- Tension within the ERM began to build up from mid-July 1992, concentrating initially on the lira, then on sterling and then on a variety of other currencies. The pressure on the Finnish Markka was so strong that it abandoned its peg with the ECU. Italy raised its interest rates, but still the lira weakened repeatedly. The Bundesbank did not cut its interest rates enough and speculation continued. The pressures on both lira and sterling became so large, that both were suspended from the ERM (and depreciated rapidly afterwards).
- The foreign exchange markets remained disturbed for the rest of the year, with a renewed outbreak of speculative pressures leading to the abandonment of Sweden's peg to the ECU, devaluation of both the Portuguese escudo and the Spanish peseta in November and the abandonment of Norway's ECU-peg in December. In January 1993, Ireland witnessed pressure (mostly because of the loss of competitiveness vis-à-vis the UK, that devalued in September 1992) and finally devalued by 10%. Germany reduced interest rates in February, March and April and the pressure was eased a bit for the currencies that had not yet been realigned.

The Mexican Peso (Tequila) crisis of 1994-95

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Mexican Finance Minister Pedro Aspe speaking in Madrid, September 1994. (IMF photo)



Michael Camdessus, IMF MD

- The Mexican peso crisis was a currency crisis sparked by the Mexican government's sudden devaluation of the Mexican peso against US dollar in December 1994. The crisis was first of its kind ignited by capital flight (outflows) and was also the first case of market-led contagion amongst EM countries
- Pedro Aspe, Mexican FM, was a star before the crisis hit. In September 1994 at mid-night in Madrid, he asserted convincingly, had been a “pioneer” in transforming its economy from one that was closed to competition, highly controlled and regulated, and wracked by high debt and inflation, into a modern, open, and efficient economic system based on market principles. His audience included IMF MD, Michel Camdessus, Lewis Preston, the World Bank President, Wim Duisenberg, the respected head of the Netherlands central bank, the governors of the Bank for International Settlements (BIS) and renowned economist Jacques Polak, who spoke on behalf of the Bretton Woods conference grouping. Pedro Aspe was PhD student of Stanley Fischer, an MIT professor.

Tequila crisis of 1994-95

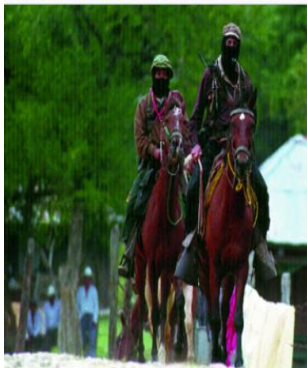
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Stanley Fischer, IMF First DMD



Colosio's
monument in
Mexico city



Zapatista Army of National
Liberation (EZLN) insurgents in
Mexico.

- Stanley Fischer moved to IMF around the same time as the First deputy Managing Director. On December 19, 1994 at 11:00 PM he was talking to Jeffrey R. Shafer (Assistant Secretary for International Affairs at the U.S. Treasury) about financial developments in the Middle East. Shafer dropped a bombshell in the passing. By the way, Shafer injected, you know that Mexico is going to devalue and float its peso tomorrow. Fischer did not know.
- During the 1994 Presidential elections, the incumbent administration embarked on an expansionary fiscal and monetary policy. The Mexican Treasury began issuing large amounts of short-term debt instruments denominated in Mexican pesos with payments indexed to US dollar (***Tesabonos***), attracting foreign investors. Mexico's access to international capital markets had soared following formation of NAFTA.
- However, a violent uprising in the state of Chiapas and the assassination of the PRI presidential candidate Luis Donaldo Colosio Murieta on March 23, 1994 had resulted in political instability, causing investors to place an increased risk premium on Mexican assets.

Lessons from the Tequila crisis of 1994-95

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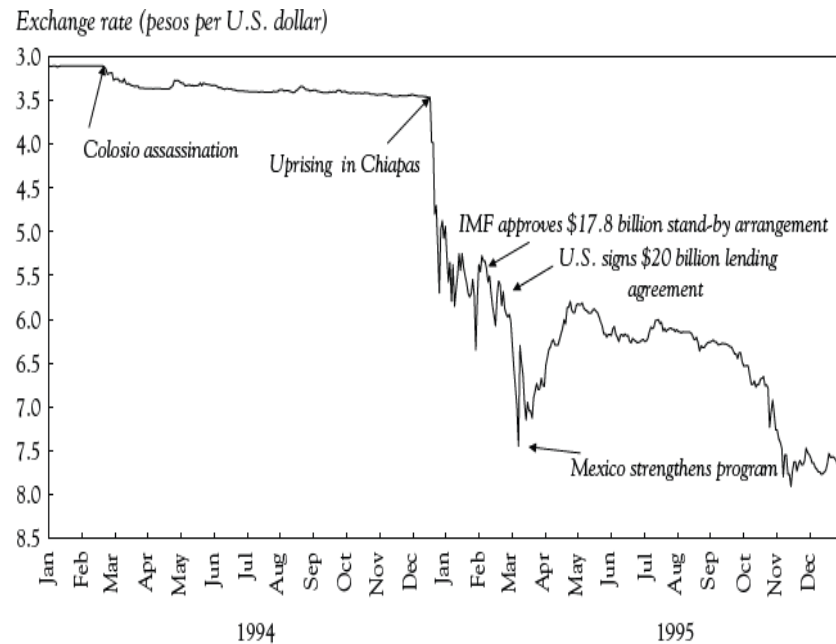


President Ernesto Zedillo

- In response, the Mexican central bank intervened in the foreign exchange markets to maintain the Mexican peso's peg to the U.S. dollar by issuing more Tesabonos & Cetes (bills). The peso's strength caused demand for imports to increase in Mexico, resulting in a trade deficit. Speculators recognized an overvalued peso and capital began flowing out of Mexico to the United States, increasing downward market pressure on the peso. Under election pressures, Mexico purchased its own treasury securities to maintain its money supply and avert rising interest rates, drawing down the bank's dollar reserves. Supporting the money supply by buying more dollar-denominated debt while simultaneously honoring such debt depleted the bank's reserves by the end of 1994.
- On December 20, 1994, the new administration of President Ernesto Zedillo Ponce de Len broadened the band in which the Mexican peso was allowed to float against the U.S. dollar, in effect permitting a peso devaluation of 15%. The market reacted by intensifying speculative attacks on Mexico's currency. Hard-currency reserves fell precipitously. Two days later, the peso was allowed to float freely against the dollar, triggering a further flight of capital, which in turn precipitated a currency crisis.

Tequila crisis of 1994-95

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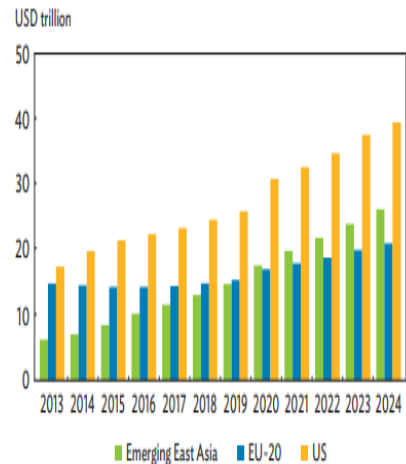
- The central bank devalued the peso on December 20, 1994, and foreign investors' fear led to an even higher risk premium. To discourage the resulting capital flight, the bank raised interest rates, but higher cost of borrowing merely hurt economic growth. Unable to sell new issues of public debt or efficiently purchase dollars with devalued pesos, Mexico faced a sovereign default.
- Two days later, the bank allowed the peso to float freely, after which it continued to depreciate. The Mexican economy experienced inflation @52% and mutual funds started liquidating Mexican assets as well as EM assets in general. The effects spread to economies in Asia and the rest of Latin America. The United States organized a \$50 billion bailout for Mexico in January 1995, administered by the IMF with support of G7 and BIS.
- The contagion from the Mexican peso crisis affected neighbouring Latin American (LatAm) countries

What is the Key Lesson from the Tequila crisis of 1994-95

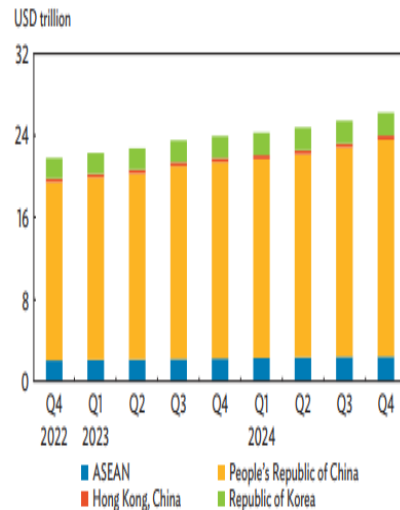
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Figure 1: Local Currency Bonds Outstanding and Issuance

A. Bond Market Size in Select Global Markets



B. Bond Market Size in Select Emerging East Asian Markets



The size of the LCY bond market in emerging East Asia reached USD26.3 trillion at the end of 2024, increasing 9.6% year-on-year (y-o-y) and surpassing the growth of 5.0% y-o-y in the United States' (US) bond market (USD39.7 trillion) and 5.1% y-o-y in the European Union 20's (EU-20) bond market (equivalent to USD21.0 trillion)

- The Key Lesson is that EMs fail to avert original sin and borrowing in foreign currencies continue to expose them to external debt crisis.
- **“Original sin”** is a term borrowed from Christian theology but has a very different context in international finance. to describe the inability of a country (typically an emerging market or developing economy) to borrow abroad in its own domestic currency. Instead, their external debt is almost always denominated in a major foreign currency, most commonly the U.S. dollar, British pound, euro, or yen.
- The term in the international finance context was proposed by Barry Eichengreen, Ricardo Hausmann & Ugo Panizza in a series of papers that refer original sin to a situation in which "most countries are not able to borrow abroad in their domestic currency.
- In the case of Mexico, Tesabonos were intended to overcome original sin by borrowing in peso but by exchange rate compensation guarantees, but the clever move boomeranged and showed the persistence of original sin